

typically use an array of sensors, tracking devices, and surveillance technology to gather information in real-time from interconnected devices. The information relevant to city operations is then centrally processed and analysed. AI-based surveillance can monitor crowds in retail stores and identify those who have potential chances of engaging in activities like shoplifting.

As systems become smart and everything is connected, they also become vulnerable to cyber-attacks. Facial recognition technology has also been incorporated in many smartphones for protecting user data from outside attacks and hacks. AI and deep learning can detect such threats quickly and are being leveraged in most cybersecurity software and products in the market these days.

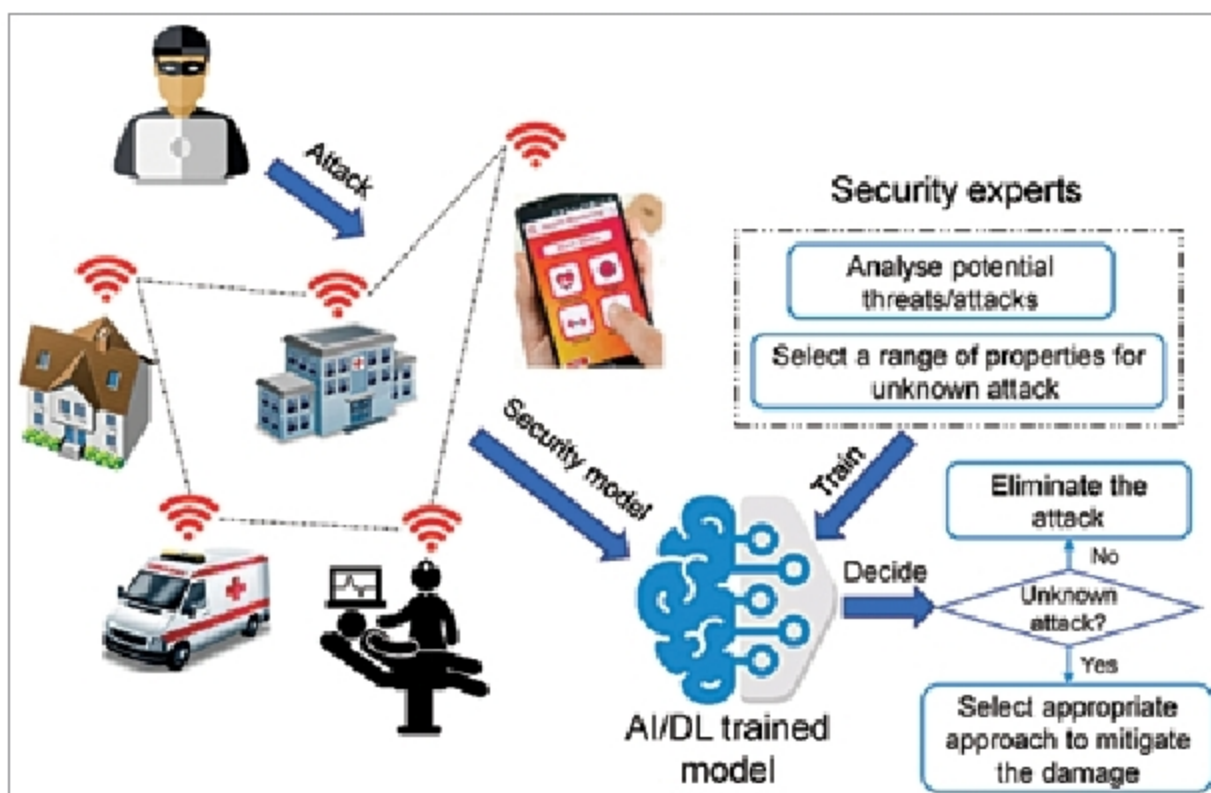
Impact of Covid-19 pandemic

Amidst the need for social distancing during Covid-19 lockdown across different countries, AI-based technological solutions that enable contactless functioning are at their peak. Startup Landing AI has created an AI-enabled social distancing detection tool that represents each individual in the video as a dot and turns bright red when the computed distance between two people is less than 1.8 metres (six feet).

In India, Vehant Technologies, a physical security and surveillance solutions manufacturer, recently launched Covid Analytics - an AI and computer vision-based intelligent image analytics solution. It detects violations through face mask detection, social distance detection, and vehicle movement detection through automatic number plate recognition (ANPR). It can be deployed both indoors and outdoors.

Enabling technologies

AI does not work in isolation to achieve its goal. Technologies like the Internet of Things (IoT) and cloud computing play a key role in the success of AI implementation in security systems. This is what makes devices like smart speakers, automotive built-in cameras and remote keyless entry locks behave as efficient tools for AI-powered surveillance. As an increas-



Use of AI in cybersecurity (Credit: www.mdpi.com)

ing number of devices get connected to each other in IoT systems, huge amounts of data need to be shared for processing. Giants like Microsoft, Alibaba, Amazon, IBM have cloud computing data centres, a more secure method of data storage and communication than traditional on-site infrastructure, for facilitating AI tasks. The data stored in cloud servers is usually accessible through the Internet. After connecting hardware to the cloud via 4G/5G networks, specialised software works to interpret this Big data collected from different locations and allows users to observe necessary measures to protect both people and assets.

Future scope

Although AI helps in spotting crimes and accidents easily, it also raises serious concerns about the future of privacy and risks to the current social structure. If companies and governments fail to take adequate steps to monitor and control the privacy violations in this field, it can have disastrous consequences. The opacity regarding the usage of data obtained in the process of surveillance varies in different nations.

Processing large amounts of information is necessary to obtain that one useful data, which implies heavy computing with multiple algorithms that require hundreds of servers. This wastage of resources leads to over-the-top expenditure on hardware. This limits its expansion in a country

like India where hardware is usually imported from other countries. When quality is often lowered to maximise storage, deep learning algorithms need to be able to evaluate these low-quality images. Also, the Internet connection needs to be faster for limited latency in processing data.

In addition, it is not always accurate or even fair, as seen from the soaring use of deep-fake videos. Take face recognition systems, for example, when the weather is poor or the images are distorted, false positives are high. Other reasons for the high false-match rates are consistent biases related to minorities and genders in AI training datasets and prejudice due to predictive policing tools. Such inconsistency is highly risky when it comes to situations of national/international importance.

In spite of the challenges, the spread of AI is undoubtedly proliferating. Innovative solutions are coming up and everyone from widely recognised security vendors to startups are contributing to these surveillance and security operations. Huawei, IBM, NEC Corporation, Hikvision, Cisco, ZTE are among the prominent companies that cater to AI surveillance necessities of most countries. There is a crucial need to not just blindly leverage the benefits of AI in surveillance but to establish a balance between AI technology, government surveillance, and the privacy rights of citizens. **EFY**